The Indemnity Ladder: How Baskets, Caps and Price Move Together in Acquisition Agreements

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Abstract

Private acquisition agreements limit the seller's liability for breached representations with a ladder of negotiated terms: a per-claim de minimis, a basket that losses must exceed before the seller pays, a cap on the seller's total exposure, and a survival period for claims. We ask whether these terms move together: comparing deals of the same price, does a higher cap go with a higher basket? We extract the ladder from tens of thousands of agreements filed with the SEC, check the extraction against 279 hand-marked agreements, and test hypotheses registered in advance on three held-out samples spanning 1995–2025. The basket rises with both the price and the cap, each holding the other fixed, and which one it follows depends on the industry. In manufacturing and services it follows the price: on 393 held-out deals, twice the price goes with a basket about 50% larger and twice the cap with one about 25% larger (elasticities 0.61 and 0.31 [0.17, 0.44]). In financial-sector deals, including real-estate trusts and insurers, it follows the cap: about 70% larger for twice the cap and 18% for twice the price (0.78 [0.67, 0.91] and 0.24; 100 deals). Both patterns were registered from a development sample. A single pooled elasticity, 0.49 [0.41, 0.58] on 637 deals, mixes the two. Within each group the basket tracks the cap more closely in larger deals, by about 0.1 per tenfold increase in price. Deals with representations-and-warranties insurance have far lower caps and somewhat lower baskets. The estimates cover deals whose basket and cap are both stated in dollars, about two-thirds of those in which we find both terms.

1 Introduction

When a buyer acquires a private company, the seller's representations about the business are backed by an indemnity, and the indemnity is bounded. Three numbers do the bounding. A *de minimis* excludes individual claims below a small amount. A *basket* (also called a deductible or threshold) requires aggregate losses to exceed a larger amount before the seller owes anything; once it is crossed the seller pays either from the first dollar (a "tipping" basket) or only the excess (a "deductible"). A *cap* limits the seller's total exposure. Together they form a ladder, and every rung is a negotiated number. Figure 1 shows one.

The theory of contract design suggests the rungs should not be independent. Indemnity terms are the instruments through which parties allocate the risk created by information the seller has and the buyer does not (Gilson, 1984), acquisition parties use even vagueness strategically (Choi and Triantis, 2009), and bargaining power can shape non-price terms, not just price (Choi and Triantis, 2012). If so—and the co-movement is our inference, not theirs—the terms should vary together across deals, and a term is best understood alongside the others in the same agreement.

This paper measures how the terms vary together. It makes three contributions.

A measurement of the ladder at scale. Generic extractors find dollar amounts in legal text (Bommarito II et al., 2021); the hard problem is deciding *which* rung an amount is. A single sentence can state a per-claim minimum, an aggregate threshold and a cap, and the same words ("exceeds $500,000") mean a threshold in one clause and a ceiling in another. We build a rule-based reader that assigns each amount its role from its context, apply it to tens of thousands of SEC-filed agreements, and validate it against 279 agreements marked by hand, against the agreements' own defined terms, against a large-language-model baseline, and against published deal-point medians.

Pre-registered estimates of how the rungs co-move. Across deals, the basket rises with both the price and the cap, each measured holding the other fixed, and which one it follows more depends on the industry. In manufacturing and services the basket follows the price more than the cap (elasticities about 0.6 and 0.3); in financial-sector deals it follows the cap (about 0.8 and 0.25). Both orderings were registered from development and hold on the pooled held-out deals. A single pooled elasticity, 0.49, mixes the two and is reported as a summary, not as the finding. The pooled estimate replicates on two held-out reads drawn different ways across two eras; on a third, of 78 deals, the estimate is lower and its interval too wide to meet the registered rule, and the relationship is a gradient in deal size rather than a constant. All hypotheses, decision rules and the order in which data were opened were registered before the relevant numbers existed (Nosek et al., 2018); deviations are listed in the appendix.

A test of what "reading the context" buys. A pre-registered ablation removes the reader's context one rule at a time. Three rules carry the extraction: distinguishing ceilings from thresholds, per-claim from aggregate scope, and reading the agreement's own name for an amount. About 400 characters of text on either side of an amount are enough. How much the rules buy over a keyword matcher depends on the truth they are scored against, and the two truths we have are biased in opposite directions. Of the six rules, only scope moves the headline appreciably: removing it lowers the elasticity by about 0.04, a shift that is clear only in financial deals (about 0.08). The sign, the distance from the null and the ordering of the regimes survive every simplified reader.

Section 2 walks through an indemnity clause, the question we ask of it, what the extractor does to it, and what the data can and cannot see. Section 3 places the work. Section 4 describes the corpus and the population. Section 5 describes and validates the extractor. Section 6 sets out the registered design, Section 7 the results, Section 8 the context ablation, and Section 9 what the results explain and where they stop.

2 The indemnity clause

2.1 What the clause says

In a private acquisition the seller makes representations about the business: assurances to the buyer that what they are paying for is as described. If one proves false after closing, the buyer's remedy is the indemnity; The seller repays the buyer's losses. The indemnity is limited, and Figure 1 shows how, in the article of a 1998 asset acquisition agreement filed with the SEC.

Four terms bound this seller's exposure. The *survival period* gives the buyer twelve months after closing to bring a claim. The *de minimis* ($5,000) discards any single claim at or below that size; a $4,000 problem is the buyer's own. The *basket* ($50,000) is a threshold on losses added together: until qualifying claims total more than $50,000 the seller pays nothing, and here it then pays "only for the amount in excess," so the first $50,000 stays with the buyer. A basket that works this way

10.1 Survival of Representations. The representations and warranties made by the parties in Articles IV and V of this Agreement shall survive the Closing for a period of twelve (12) months [survival], except for the representations set forth in Section 4.15 hereof which shall survive for a period of thirty-six (36) months ...

10.2 Indemnification by Seller. Seller shall indemnify and save harmless Purchaser ... from and against any and all losses, liabilities, claims, ... damages ... ("Claims") ... In no event shall the aggregate obligation of Seller under this Section 10.2 exceed $2,250,000 [cap]. Notwithstanding anything to the contrary set forth in this Agreement ..., Seller shall not be liable hereunder to Purchaser as a result of any breach of any representation or warranty contained in this Agreement (i) for any individual Claim that does not exceed $5,000 [de minimis] and (ii) unless and until the Claims incurred by the Purchaser ..., in the aggregate, as a result of any breaches exceeding $50,000 [basket], and then only for the amount in excess of $50,000.

Figure 1: An indemnity clause: Sections 10.1–10.2 of a 1998 asset acquisition agreement filed with the SEC (EDGAR accession 0000891618-98-000114), abridged where marked .... Shaded amounts are the terms; underlined words are the cues that decide each amount's role.

is a *deductible*; one that, once crossed, makes the seller pay from the first dollar is a *tipping* basket. The *cap* ($2.25 million) is the most the seller can be made to pay in total. Against the $7 million price, the basket is 0.7% and the cap 32%. Figure 2 sets the terms against each other on a dollar scale.

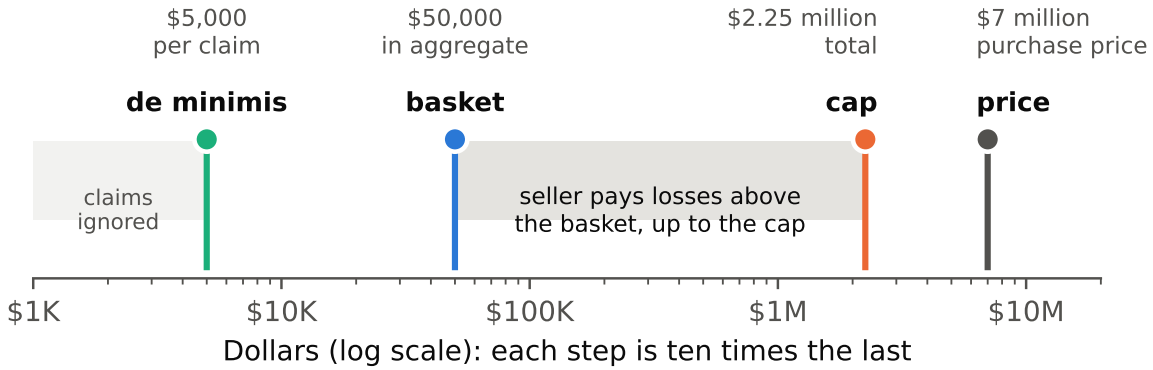


Figure 2: The terms of Figure 1 on a dollar scale, with the purchase price ($5 million in cash and $2 million in stock). The survival period is a length of time and is not shown.

2.2 The question

We ask if these terms move together across deals. Does a deal with a higher cap also tends to have a higher basket, and if that holds when the deals compared are the same size. There are reasons to expect it. A seller who concedes a high cap may reject a high basket; both terms may reflect how risky the business is; or a law firm's template may have given acceptible ranges for both. The data cannot tell these apart, but they can say whether and how strongly the terms move together.

Figure 3 shows the five relationships we registered: the triangle of basket, cap and price, and

two edges to the outer terms, the de minimis to the basket and the survival period to the cap. The main estimate is the basket–cap edge, measured as an *elasticity*: the percentage change in the basket that goes with a one-percent change in the cap, comparing deals of the same price. An elasticity of 0.3 means that, at the same price, a deal with twice the cap has a basket about 23% larger ($2^{0.3} \approx 1.23$); an elasticity of 0.75 means about 68% larger; zero means the cap tells you nothing about the basket.

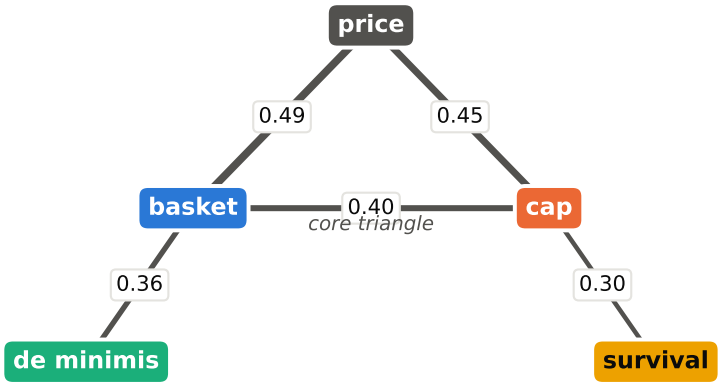


Figure 3: The five relationships the study registered in advance. Each number is a partial correlation in the development data: the association between two terms with the other terms held fixed, where 0 is none and 1 is perfect. All five exclude zero in development; how they fare on held-out data is in Table 3. The paper's main estimate is the basket–cap edge at fixed price.

2.3 Reading the numbers

A filed acquisition agreement runs to dozens of pages, and its indemnity article often states several dollar amounts besides the terms: escrows, special-matter caps, the buyer's own limits. In Figure 1 the word "exceed" introduces all three amounts—the cap, the de minimis and the basket—so a tool that labels an amount by the nearest keyword cannot tell them apart. Our extractor reads each amount's role from the words around it, in the steps of Table 1: is the amount a ceiling or a threshold, does it apply per claim or to all claims together, does the agreement give it a name, is it tied to a specific matter (a tax or environmental claim, an escrow) and so not a general term at all, and does it limit the seller or the buyer. This agreement has a second $50,000 basket in Section 10.3 that limits the *buyer's* indemnity of the seller; the extractor records it and leaves it out.

Run over tens of thousands of agreements, the extractor gives one ladder per deal; a language model reads the purchase price separately (Section 4). The first author checked the extractor by hand on 279 agreements (Section 5).

2.4 What the data is: agreements that were signed

Every agreement here is a final, signed contract that a public company filed with the SEC because the deal was material to it. We see the terms the parties settled on, not the drafts, the markups traded between counsel, or the terms each side asked for and gave up: a cap that opened at half the price and closed at 15% appears only as 15%. Deals that broke down before signing leave no trace, and neither do deals no one had to disclose—acquisitions between private companies, and deals too small to matter to a public filer. A filed agreement records a signing, and some signed

Table 1: How the extractor assigns roles in the clause of Figure 1. A keyword matcher sees "exceed" beside all three amounts.

Amount	Words that decide it	Ceiling or threshold?	Per claim or total?	Role
$2,250,000	"In no event shall the aggregate obligation … exceed"	ceiling	total	cap
$5,000	"for any individual Claim that does not exceed"	threshold	per claim	de minimis
$50,000	"unless and until … in the aggregate … exceeding"	threshold	total	basket
$50,000 (again)	"then only for the amount in excess of"	how the basket pays	–	deductible
twelve months	"shall survive the Closing for a period of"	–	–	survival period
$50,000 in Sec. 10.3	heading: "Indemnification by Purchaser and Sub"	threshold	total	buyer's basket; not used

deals never closed. The results describe the terms of deals that were agreed on, compared with each other. They do not describe the course of a negotiation, and the agreed deals need not be typical of the negotiations behind them: if deals whose terms fit poorly together were more likely to fall apart, the signed agreements would show a tighter relationship than negotiators face at the outset. This data cannot measure that.

3 Related work

Indemnification in M&A practice. Coates (2012) documents the allocation of risk through M&A contracts, including the clustering of baskets on round numbers and on whole or half percentages of deal value; Coates (2016) shows that target indemnification appears in most private-target deals and few public-target ones, which is why this paper's population is private-target acquisitions filed as exhibits. Anderson IV and Manns (2017) argue that merger agreements evolve inefficiently, through template churn rather than deal-specific drafting, a conclusion others contest (Nyarko, 2021). Nyarko (2021) shows that whether a contract contains a dispute-resolution clause is driven almost entirely by the template supplying the first draft; we extend the question from the *presence* of a term to its *magnitude*. Griffith (2019) documents how representations-and-warranties insurance (RWI) changes indemnity terms, shrinking the seller's exposure to a fraction of its uninsured level.

Contract element extraction. Chalkidis et al. (2017) framed contracts as sets of typed elements to be located and extracted, and Chalkidis et al. (2021) found that recurrent (BiLSTM) encoders outperformed dilated CNNs, transformers and BERT on the task, which they attribute to the lack of recurrence in transformers on a task whose entities are highly context-sensitive. On SEC filings specifically, subword tokenization fragments numbers and hurts numeric tagging, though domain pre-training and numeric pseudo-tokens close much of the gap (Loukas et al., 2022). Domain pre-training improves legal classification and contract-element tagging (Chalkidis et al., 2020), and classification benchmarks (Chalkidis et al., 2022; Tuggener et al., 2020) address provision *type*; none assigns an amount to its role in an indemnity ladder. CUAD (Hendrycks et al., 2021) annotates spans for 41 clause categories, including caps on liability but no indemnification category, and does

not assign amounts to roles; ContractNLI (Koreeda and Manning, 2021) is a different task. The closest prior dataset in domain, MAUD (Wang et al., 2023), is built on the ABA's public-target deal-points study and poses no questions on indemnification, baskets, caps, de minimis or survival—unsurprisingly, since post-closing indemnification is rare in public-target deals (Coates, 2016). The closest prior system, LexNLP (Bommarito II et al., 2021), extracts amounts but does not say which rung an amount is.

Corpora. Recent corpus papers assemble SEC-filed contracts at scale (Adelson and Nyarko, 2025; Adelson et al., 2025) without extracting indemnity terms. To our knowledge no prior work assigns extracted amounts to indemnity roles at this scale and studies their joint distribution; we state this as a claimed gap from our search, not an established fact.

4 Data

4.1 Corpus

The source is a local archive of 1,117,867 EDGAR exhibit documents (421,664 accessions submitted under 8,337 CIKs, many of them filing agents) in the format of the KL3M data project (Bommarito II et al., 2025), from which a contract corpus of 253,936 documents was rebuilt. A document classifier labels a subset as acquisition or purchase agreements (acquisition agreements it misses are largely recovered by the archive test, (iii) below); restricted to exhibits whose text marks them as material contracts (EX-10) or plans of acquisition (EX-2), that subset is 61,214 documents, split once by submitting CIK into a development set of 47,543 and a holdout of 13,671 that no probe, label or hand mark touched before it was read. Because many documents are submitted by filing agents, 77 of the 435 holdout core deals come from companies also present in development; the other 358 come from companies development never saw. No holdout agreement is a copy of a development agreement: the holdout is screened against development by the text of the indemnity article (386 gated documents removed). The sandbox is overwhelmingly 1995–2020 filings.[1]

Held-out material. Three further samples were built for testing. (i) *Post-2019 filings*: EX-10 and EX-2 exhibits filed 2020–2025, pulled from EDGAR full-text search, with no overlap with the sandbox, split by filer into development, test-A and test-B under a registered seed; the split was drawn once, found to put 28% of the documents then pulled in one connected component, amended before any outcome existed, and frozen by hash. Three registered waves pulled 35,761 exhibits. The search returns at most 100 hits per query and month, so a fourth pull recovered 18,066 exhibits it had dropped (53,827 in all), each assigned by the registered extension rule without moving any earlier assignment. Test-A was opened once. (ii) *Test-B and test-A additions* were opened together as a "joint" read of 2020–2025 deals; the recovery enlarged test-B, so the joint read reported here is a second reading of it (Appendix A). (iii) An *archive test*: 1995–2020 agreements found by reading all 1.1 million archive documents for indemnity language outside the sandbox, never seen by development. No held-out sample contains an agreement that development, or another held-out sample, already has: each is screened by the text of the indemnity article, which leaves 3,316 gated archive documents.

[1]The archive's manifest dates 936 sandbox documents (1.5%) 2022–2025. They barely reach the regressions: four development core deals and no holdout core deal are dated after 2020, and one of the 417 archive-test deals. "1995–2020" names the era of the sandbox core.

4.2 From documents to deals

Deal gate. Indemnity clauses also appear in leases, licences and service contracts. A clause-type score separates deal indemnity from operational indemnity; its threshold was calibrated against 640 usable excerpts labelled by Claude Opus 5; on the score the gate applies, the share labelled deal indemnity rises from 72% at a score of 5 to 80% at 6 and 89% at 7, and 94% of the documents it admits are deal indemnities. The excerpts were drawn from the start of each development shard, so they over-represent early filings (median year 2000, against 2007 in development; 6% filed from 2015 on, against 27%). Among labelled documents the gate admits, the share that are deal indemnities is stable across eras: 93% before 2005, 95% in 2005–14 and 100% from 2015, where few labels fall. In development the gate keeps 13,941 of 47,543 documents.

One record per negotiation. The same agreement is often filed several times, by both parties or in successive reports. Near-duplicates are collapsed with a 32-hash MinHash over five-word shingles of the indemnity region (Broder, 1997) at estimated Jaccard similarity ≥ 0.8 when the primary rungs also agree (1,830 documents removed in development). Records that repeat a ladder are kept but weighted $1/k$, where k is the number of the issuer's records carrying an identical ladder (every stated rung amount, any reference cap, the survival period and the basket type), so a drafting decision copied across deals counts once; every record also carries an EDGAR issuer cluster for inference. In practice k is small: 2.1% of the development core sits in such a group, the largest has three deals, and most groups also share the purchase price (11 of 18), so the weight chiefly catches the same deal filed twice that the text filter missed—sometimes as two exhibits of a single filing. Where the price differs, the group can again sit inside one filing: several exhibits that are companion agreements of one transaction, each with its own price but one negotiated ladder. The weighting does not move the results.[2]

Price. The purchase price is labelled by a language model (Claude Sonnet 5) from excerpts of each agreement: value, currency, kind (stated, base, formula, per-share), escrow, and whether the consideration includes a component with no stated dollar value. Only whole US-dollar prices are used: a price with an unvalued component is a floor, not a price, and is never divided into a rung. A blind hand check by the first author of 50 held-out deals agreed exactly on 41 of the 44 both priced (Appendix B). All language-model labelling on the project cost about $130 in total.

Error in the rungs. Extraction error does not inflate the estimate. On 173 hand-checked deals, the elasticity computed from the true values is within about 0.03 of the one computed from the extracted values, and the small difference runs toward zero, not away from it (Appendix B).

The core. The regressions use deals with a dollar basket, a dollar cap and a whole USD price, with the basket below the cap, the cap at or below the price, and no rung under $1,000. In development the core is 2,266 deals; in the holdout 435.

Scope. We measure each agreement as a snapshot of initial liability at signing. Some agreements schedule their terms to change; caps that step down on an anniversary or on escrow release, caps tied to stock consideration, caps tiered by the class of representation breached. We set these aside

[2]Weighting every deal equally, weighting $1/k$, and dropping deals with $k \geq 2$ give cap coefficients of 0.41, 0.41 and 0.41 in development (2,266 deals), 0.40 in all three arms in the holdout, and 0.50, 0.49 and 0.49 on the 637 pooled held-out deals; no price coefficient moves by more than 0.01 (exploratory).

deliberately: a snapshot gives every agreement one comparable value per term. Of the 4,689 priced development agreements with a stated cap, 1.1% have a cap that steps down over time (read by hand from regex candidates; a floor, since only single-sentence schedules are found), and about 5% treat fundamental, title or tax representations under a different cap, rising from almost none before 2005 to about one in eight from 2015 (regex, 72–84% precision on a hand-read sample).

Population boundary. The core admits only terms stated in dollars, for the basket and the cap alike, and it was defined that way before any held-out read. A term stated by reference—"the Purchase Price", "the Escrow Amount", "10% of the Purchase Price"—has to be resolved against a labelled price or escrow, a second reading step with its own errors. Of development deals with a basket and a cap, the reader finds the cap stated in dollars in 67.0%, as the whole purchase price in 21.7%, as the escrow in 8.4% and as a percentage of the price in 2.9%. The percentage share is an undercount, because the reader misses percentages written in words. The whole-price share is an overcount. The first author marked 25 of these agreements drawn at random: in 13 the price caps the seller's general liability; in 12 it does not. Most often the price is only the ceiling for fundamental representations, while the general cap is a dollar amount or a percentage of the price the reader missed. Deals truly capped at the whole price are about 11% (95% interval 7–15%). Our estimates describe deals whose terms are stated in dollars.

5 Measuring the ladder

5.1 The reader

The extractor first finds the indemnification region; the article headed indemnification, survival or limitations, with a text-density fallback when headings fail (86% coverage of labelled rungs). Within it, every dollar amount is assigned a role by the steps of Section 2.3: direction (a ceiling or a threshold), read from the nearest direction cue within 200 characters before the amount or, failing one, an explicit ceiling phrase earlier in the same sentence; scope (per claim or aggregate); the agreement's own name for the amount; carve-outs (amounts tied to a specific matter, escrow amounts and price adjustments are recorded but never seated as rungs); and party, read first from the sentence itself (which side is named as liable or as indemnified, with the parties' names learned from the indemnity article's grants and the preamble), then from a section the sentence cites ("under Section 9.2"), and failing both from the heading of the subsection it sits in. Caps stated against a named amount ("capped at the Purchase Price", "10% of the Purchase Price") are recorded as reference caps and resolved to dollars with the labelled price; they are read, but they are outside the core (Section 4.2). The extractor was frozen before the held-out samples were opened. Two later readers are reported in Appendix C: one for percentage baskets and one for percentage caps.

5.2 Validation

Hand-marked agreements. The first author marked 279 agreements on a purpose-built page that shows each extracted rung in context, blind to the analysis. Cards were drawn in strata by the residual of the basket–price relationship, plus a stratum of agreements where the extractor bound no basket or no cap. Two hundred of the cards are agreements on which a 2026 revision of the reader's rules was developed (after a model audit flagged errors on them); precision on those is in-sample. The remaining cards were drawn afterwards and never used to write a rule; 28 of them form a fresh band of moderate residuals. Table 2 reports precision on the core, weighting each stratum, in-sample or fresh, by its share of core deals, and on the fresh band alone.

Table 2: Extraction precision against the first author's marks. The core rows place each card still in the core in its residual stratum on the current core and weight each stratum by its share of core deals, so they combine in-sample cards (those the 2026 rule revision was developed on) with fresh ones. The fresh rows are the fresh band alone: cards drawn after the revision from the moderate residuals, likely harder than the bulk of the core. The "re-read" rows count as extractor errors the marks overturned when the disagreements with a blind model read were adjudicated (Section 9).

	Basket	Cap	Both	Price
Core, stratum-weighted (five strata, 188 cards)	0.971	0.987	0.958	0.983
Fresh, moderate-residual band (28)	0.89 ± 0.06	0.96 ± 0.04	0.86 ± 0.07	–
Core, after re-read	0.962	0.966	0.933	–
Fresh, after re-read	0.86	0.89	0.79	–

Recall and what is missed. On 50 agreements where the extractor found no basket or no cap, the first author found a basket in 31% of the basket-silent records and a cap in 40% of the cap-silent ones, which puts basket recall over the priced set at no less than 0.86. The misses are not random: A basket stated as a percentage of the price ("1% of the Purchase Price") appears in 23.8% of the regions where the rules bound no dollar basket, against 4.0% where both rungs were bound; four of the eight baskets the first author found on silent cards were percentage-stated. Most such baskets belong to deals whose cap is also stated by reference, outside the core.

A language-model baseline. Given the same text, a language model (Claude Sonnet 5.5) came close to the rules on amounts in the core (basket 0.966 against 0.971, cap 0.938 against 0.967), was ahead where the rules had missed a rung, and was much better at whether a basket tips or deducts (95% agreement with the first author against 81%), a call that often depends on the sentence after the threshold. The model's basket-type labels exist only for the marked agreements; where basket type enters the analysis (the template weights and the drafter covariates of Section 7.5), it is the rules' label (Appendix B).

Published benchmarks. We compare each edition of the American Bar Association's Private Target M&A Deal Points Study (American Bar Association, Business Law Section, Mergers and Acquisitions Committee, 2006–2025), which codes publicly filed agreements for the sale of privately held companies (the same filings and deal type as our core), with our deals from the same years and size band: ten editions, 2006–2025, each covering one or two years of deals. The two are different kinds of sample. Each edition hand-codes 100–151 agreements selected within a band of fairly large transactions (from $17–50 million up to $500 million–$4.7 billion, varying by edition); our comparison deals are every filed agreement of those years in the same band, 112–242 per edition, and our core as a whole spans deals of every size. The median cap is within 1.1 points of the study's in every edition through 2014. From 2016 the study's overall median falls as insured deals with nominal caps become common; among agreements that do not mention insurance, ours stays within 1.8 points (8.6–10.0% against 8.4–11.8%). The median basket is within 0.1 points of the study's in six of the ten editions. Our share of baskets at or under 1% of price is lower in all but one edition (75–91% against 88–97%), so our right tail is fatter; the extractor's basket precision on the hand-marked agreements whose basket sits far above the basket–price relationship (0.95, 23 agreements) suggests that this is the population, not misreading. Our insured share is lower (16–46% against 29–65% from 2016), which traces to population composition rather than to missed insurance flags. Figure 4 shows the comparison edition by edition.

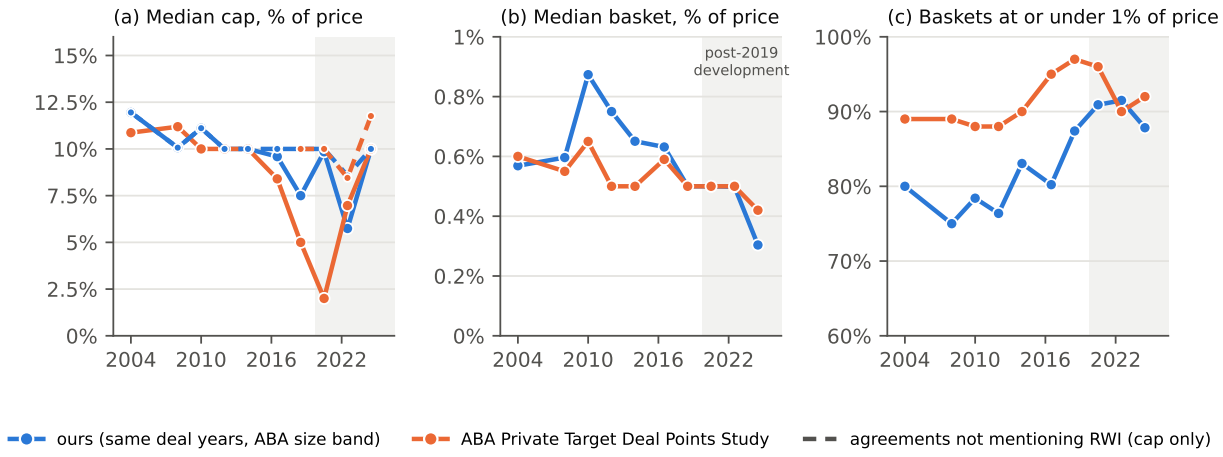


Figure 4: Our deals against the ABA Private Target Deal Points Study, edition by edition, each point at the midpoint of the study's deal years and restricted to its size band. Ours: development deals (1995–2020 sandbox to 2019, post-2019 development after, shaded; never a test set). Dashed in (a): agreements that do not mention representations-and-warranties insurance, which the study reports from 2016–17. The 2004 study prints no median cap; its value is from the 2025 study's historical series. ABA figures with their pages: docs/aba_primary.json.

6 Design

Model. The headline specification regresses log basket on log cap and log price,

$$\log_{10} B_i = \alpha + \beta \log_{10} C_i + \gamma \log_{10} P_i + \varepsilon_i, \tag{1}$$

with template weights and a bootstrap that resamples issuer clusters (Cameron et al., 2008). β is the elasticity of the basket to the cap at fixed price (Section 2), and γ the elasticity to the price at fixed cap. Figure 5 draws the equation without the algebra.

A five-edge partial correlation structure (basket–cap, basket–price, cap–price, basket–de minimis, cap–survival period) is tested alongside.

The truncation null. The core requires $\$1{,}000 \le B < C \le P$. The bound $B < C$ by itself produces a positive β, even if baskets and caps were unrelated. We measure how much by simulating baskets independent of the cap at fixed price, keeping the bounds, and refitting: the null elasticity is about 0.1 (0.11–0.12 in development, 0.11–0.13 on the pooled held-out deals, 0.07–0.08 in manufacturing and services and 0.16–0.22 in finance; on test-A's 78 deals it reaches 0.24–0.34 at the upper bound). The registered null, 0.08, came from an earlier construction. A truncated-regression estimator that models the bound directly gives estimates within 0.03 of least squares in every sample (Appendix B).

Registration and order of opening. Development used the 1995–2020 sandbox. Before each held-out sample was opened, its hypotheses, decision rules and minimum sample sizes were written into a pre-registration with a timestamped commit: for β, "REPLICATED" if the interval covers the development value 0.42 and excludes the null 0.08. Test-A was opened first, then test-B together with test-A's late additions, then the archive test. Subgroup hypotheses—industry regimes, a size gradient, industry's share of the spread—were registered from development before any subgroup

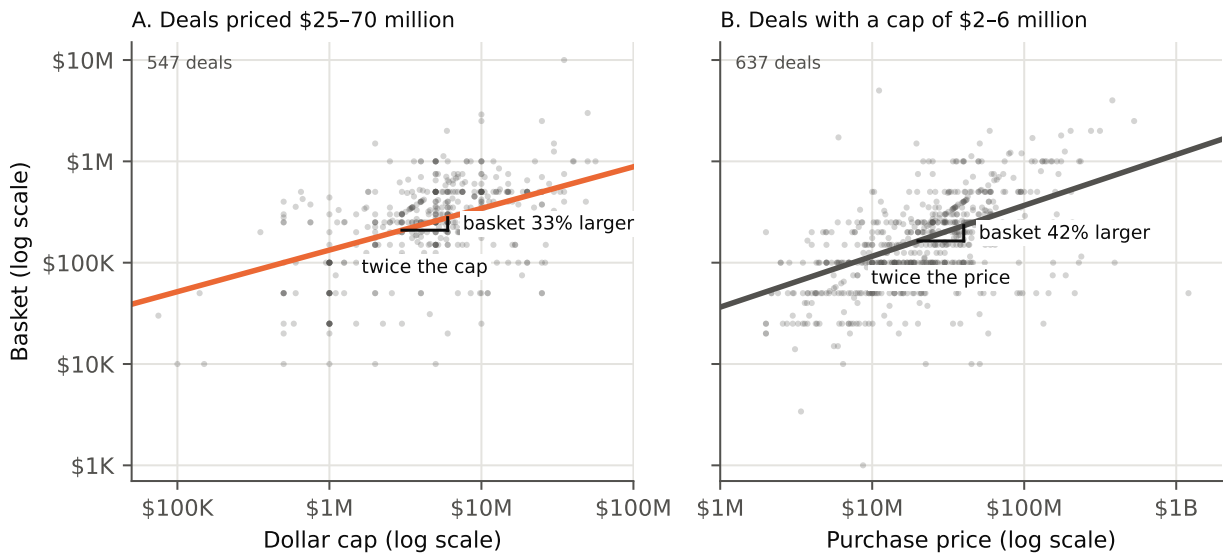


Figure 5: The equation in pictures, on the development deals. Each dot is one agreement. Panel A holds the price roughly fixed by taking only deals priced $25–70 million: among them, deals with larger caps have larger baskets, and the line's slope is β. Twice the cap goes with a basket about a third larger. Panel B holds the cap roughly fixed ($2–6 million): deals at higher prices have larger baskets, and the slope is γ. Twice the price goes with a basket about 40% larger. The lines are the full model's estimates (0.41 and 0.50 on all 2,266 deals), not fits to the panels; the scatter around them is the variation the terms leave unexplained. Both axes are on a log scale, so equal distances are equal ratios.

number existed on a test set. The 1995–2020 holdout, unlike the later samples, was read several times after rule changes justified on development data; we treat it as supporting evidence rather than a clean test.

7 Results

7.1 Two regimes by industry

In financial-sector deals (SIC 60–67) the basket follows the cap: on the pooled held-out deals the cap coefficient is 0.78 [0.67, 0.91] and the price coefficient 0.24, a difference of +0.54 [0.33, 0.76] on 100 deals. In manufacturing and services (SIC 20–39 and 70–89; "operating" in the registration) the reverse holds: price 0.61, cap 0.31 [0.17, 0.44], a difference of +0.31 [0.09, 0.54] on 393 deals. Industry is the code the filing company carried when it filed. Development, where the regimes were found, gives the same picture (financial cap 0.72, price 0.22, on 253 deals; manufacturing and services cap 0.23, price 0.65, on 1,408), and neither regime is carried by token packages, deals whose dollar cap is under 1% of price (without them: financial cap 0.74, manufacturing and services cap 0.21). In plain terms, comparing deals of the same price, a deal with twice the cap has a basket about 25% larger in manufacturing and services and about 70% larger in finance. Comparing deals with the same cap, a deal at twice the price has a basket about 50% larger in manufacturing and services and about 18% larger in finance. Figure 6 sets the two elasticities side by side: in manufacturing and services the price bar is the taller in every sample, in finance the cap bar is.

The registered rule reads both directions as holding on the pool. Within divisions, manufacturing (cap 0.30, price 0.64) and services (0.30, 0.59) anchor on price, transport and utilities leans

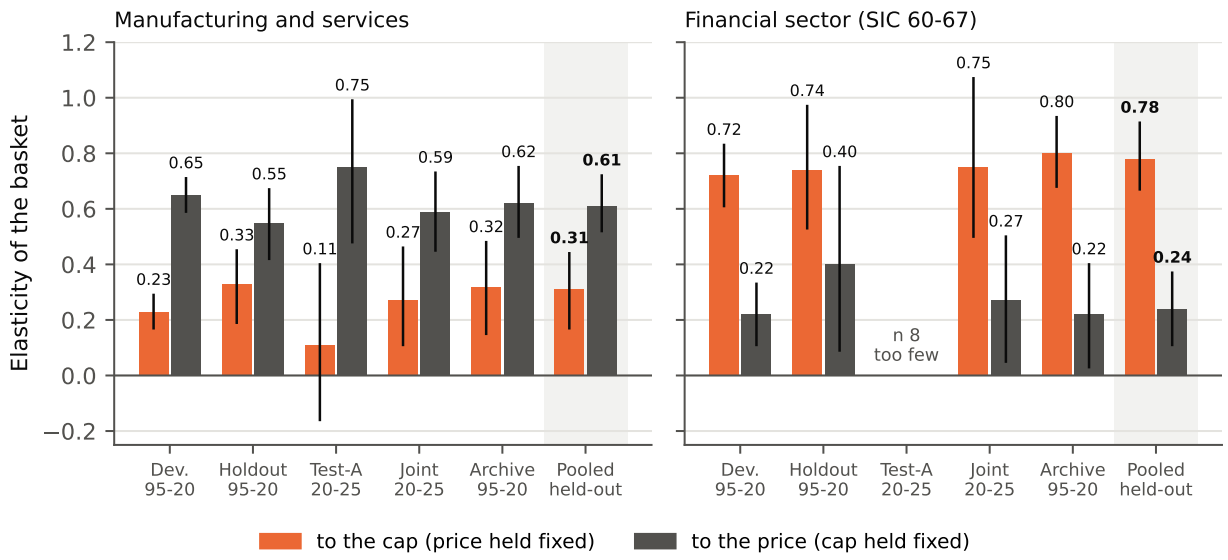


Figure 6: How the basket moves with the cap (price held fixed) and with the price (cap held fixed), by industry and sample. Bars are least-squares elasticities, whiskers issuer-cluster bootstrap 95% intervals. Shaded: the pooled held-out deals, where the registered test was read. Financial deals are too few to estimate in test-A.

toward the cap, and finance anchors on the cap. Figure 7 shows both regimes in every sample. Financial deals are too few to estimate in test-A (8); in the joint 2020–2025 read they follow the cap as elsewhere (35 deals; cap 0.75 [0.50, 1.07], price 0.27). In the two 2020–2025 reads the manufacturing-and-services coefficient is 0.11 and 0.27, with wide intervals, against 0.23 in development.

What "financial" is. Of the 100 held-out financial deals the largest group is real-estate investment trusts (44); banks are 18, insurers 17 and blank-check acquirers 1. Their caps are lower relative to price (median 8.5% against 13.3%). Four exploratory checks, not registered, rule out accounts that need no mechanism. The split is not the size gradient (Section 7.4): with the S2 term added, the financial-by-cap interaction moves from +0.47 to +0.40 [0.24, 0.57], and within the price range both groups share ($8–410 million) the cap coefficient is 0.79 for financial deals against 0.43 for manufacturing and services. It is not token packages, which are commoner in finance (financial deals without them: cap 0.78). It is not property sales: 77 of the 253 development financial deals are real-estate purchase agreements, and without them the cap coefficient (0.67) still exceeds the price coefficient, by 0.36 [0.04, 0.61] (pooled held-out 0.43 [0.01, 0.79]). And it is not one sub-sector: REITs alone give 0.69 (44 deals), the rest of finance 0.69 (56). Both are below the combined 0.78, so part of that figure is a difference between REITs and other financial firms rather than a slope within either: with a REIT intercept the financial cap coefficient is 0.69 [0.55, 0.83] and the price coefficient 0.31, still the reverse of manufacturing and services. Why these deals vary with the cap is open; the data does not test a mechanism.

The other industries. The registered comparison leaves out 27% of development deals and 23% of held-out deals: transport and utilities, mining, wholesale, retail and construction. In development these 590 deals behave like manufacturing and services (cap 0.35, price 0.58, a difference of −0.23

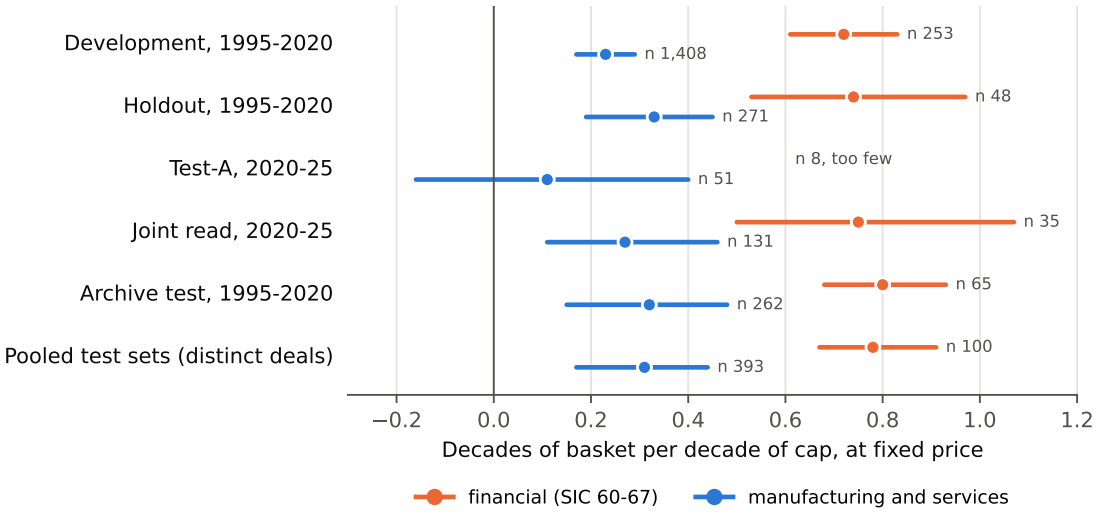


Figure 7: Elasticity of the basket to the cap at fixed price, by regime and sample, with issuer-cluster bootstrap 95% intervals (registered industry codes). Regimes with fewer than 25 deals in a sample are not estimated. The truncation nulls of Section 6 are about 0.07–0.08 for manufacturing and services and 0.16–0.22 for finance on the pooled held-out deals.

[−0.39, −0.06]). In the pooled held-out deals the 142 of them sit between the two regimes (cap 0.47 [0.30, 0.61], price 0.42, difference +0.05 [−0.30, 0.34]), and transport and utilities is the clearest intermediate case (62 deals; cap 0.55, price 0.38). The two regimes therefore describe manufacturing, services and finance; for the remaining fifth to quarter of deals the anchor is mixed on held-out data, and on development data it is the price. An earlier version of this comparison left 79 held-out deals without an industry code; with codes taken from each filing, two remain.

7.2 The pooled estimate

The regimes are the finding; the pooled elasticity β is their weighted mixture, reported because the registered tests were written on it. Without the financial deals the pooled held-out estimate falls from 0.49 to 0.35 [0.24, 0.45] and the price coefficient (0.56) exceeds it (exploratory). In development β is 0.41 [0.35, 0.46] on 2,266 deals, with the price coefficient at 0.50 and $R^2 = 0.71$. It replicates on the 220 deals of the joint 2020–2025 read (0.49 [0.34, 0.62]) and the 417 deals of the archive test (0.50 [0.39, 0.60]). Test-A, the first sample opened and the smallest, does not meet the registered rule: 0.30 [0.06, 0.55] on 78 deals covers the development value but also the null 0.08. Its point estimate is below the other samples' 0.40–0.50, and its interval, wide because the sample is small, covers all of them; and on 78 deals the conditional null itself reaches 0.24–0.34 at its upper bound (Section 6). The first read of test-A, on 86 deals that included copies of agreements development already had, gave 0.35 [0.09, 0.57] and met the rule by 0.01 (Appendix A). Pooling the 637 distinct held-out deals, the joint read and the archive test, gives 0.49 [0.41, 0.58], or 0.42 [0.33, 0.51] without token packages, which are 3.5% of these deals.[3] On filers that development never saw it is 0.43 [0.16, 0.66] (joint, 60 deals) and 0.36 [0.12, 0.55] (archive, 74). At the same price, a tenfold larger cap goes with a basket about 3.1 times larger on the pooled estimate: about 2.0 times in manufacturing and services and about 6.1 times in finance.

[3]The registered pooled read counted test-A's deals twice (759 rows); the pool here holds each agreement once. See the appendix.

Table 3: Partial correlations among the rungs (each given the other core nodes), with 95% intervals. Registered rule: an edge holds if its interval excludes zero.

Edge	Development (2,266)	Holdout[a] (435)	Test-A (78)	Joint (220)	Archive (417)
Basket–cap | price	.40 [.35,.45]	.39 [.28,.50]	.35 [.07,.58]	.49 [.35,.62]	.49 [.39,.59]
Basket–price | cap	.49 [.44,.53]	.48 [.40,.56]	.65 [.47,.78]	.46 [.33,.57]	.49 [.41,.57]
Cap–price | basket	.45 [.40,.49]	.45 [.36,.53]	.36 [.14,.57]	.41 [.27,.54]	.38 [.27,.48]
Basket–de minimis	.36 [.28,.45]	.42 [.17,.61]	n = 6	.44 [.09,.69]	.21 [−.08,.43]
Cap–survival	.30 [.23,.37]	.26 [.08,.44]	.28 [−.13,.55]	.10 [−.11,.29]	.22 [.05,.37]

[a]Supporting, not a clean test: read more than once after rule changes (deviation 3). Test-A, joint and archive were each opened once and re-read once after a code review (deviation 13).

The estimate is smaller within issuers. With issuer fixed effects the development elasticity is 0.27 [0.18, 0.35], exploratory and on development data only. The co-movement is therefore partly a between-issuer phenomenon—issuers who write larger caps also write larger baskets—and partly within: inside families of near-identical text the elasticity is 0.33–0.39.

Structure. Table 3 gives the partial correlation structure. The triangle of basket, cap and price holds in every sample. The de minimis edge is unsettled (it fails on the archive and is unmeasurable on test-A), and the cap–survival edge holds before 2020 but fails in both post-2019 reads. Test-A fails two edges, which by the registered rule means the five-edge structure does not transfer to that sample; the core triangle does. Not registered but consistent across both later reads, the cap also tracks the escrow (0.34 joint, 0.35 archive).

7.3 Levels and their drift

In development the median basket is 0.75% of the price (interquartile range 0.40–1.25%) and the median dollar cap 12.5% (5.6–25.4%). The cap has fallen; the basket has barely moved. Across five-year filing windows the median cap falls from 21% of the price in 1995–99 to 10% in 2015–19, while the median basket stays between 0.70 and 0.83%, so the median ratio of cap to basket falls from 25 to 15. Held-out deals follow the same line for the cap: in 2020–26 the median cap is 10.0% and the ratio 15, and the median basket, 0.62%, is a little lower. Across all years the held-out medians are similar to development (basket 0.50–0.75%, cap 10.0–11.1%). Figure 8 shows the drift by filing era.

7.4 A gradient in deal size

The elasticity is not a constant. Adding an interaction of log cap with log price, the cap slope rises by +0.10 [0.05, 0.15] per tenfold increase in price on the pooled held-out deals (development +0.12), holding on both adequately powered reads. Implied at development values, the elasticity is about 0.31 for a $10 million deal, 0.43 at $100 million and 0.55 at $1 billion. At the median development deal ($41 million) the interaction model gives 0.38 [0.33, 0.43], a little below the 0.41 of the model without the interaction; the gradient is the more useful summary.

The gradient is not an artefact of mixing the regimes. Financial deals are larger (median price $78 million against $35 million in manufacturing and services on the pooled held-out deals) and have the steeper cap slope, so blending alone could tilt the pooled slope with size. It does not (Table 4): the interaction is positive inside each regime, with intervals that exclude zero in both samples (in

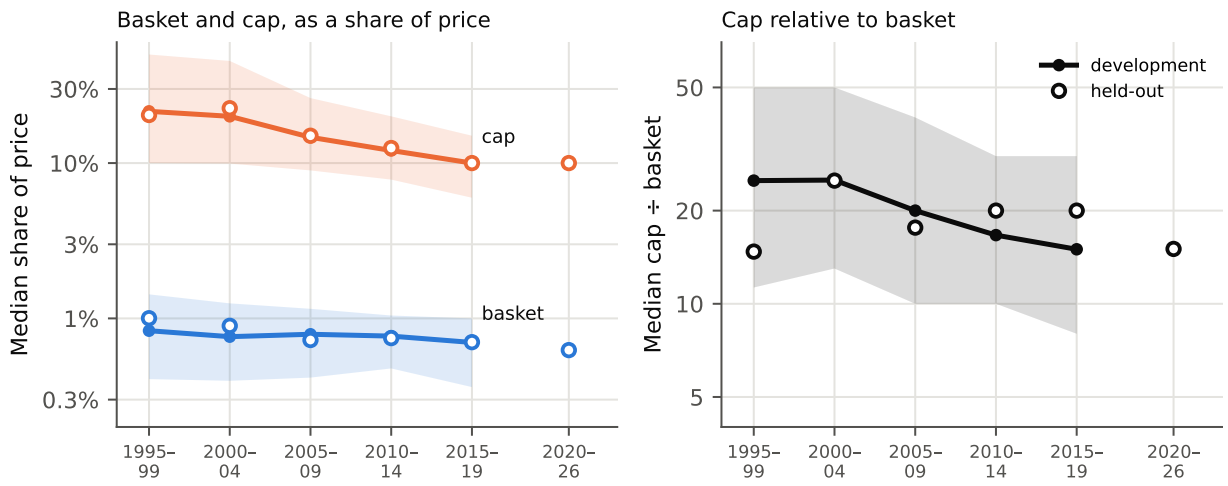


Figure 8: Levels by filing era. Left: the median basket and dollar cap as a share of the purchase price. Right: the median ratio of cap to basket. Lines and shaded bands are the development medians and interquartile ranges; open circles are the held-out medians. Five-year windows (the last, 2020–26, seven years) with at least 40 deals; both axes are on a log scale. Descriptive.

Table 4: The size gradient by regime: change in the cap elasticity per tenfold increase in price (the log cap × centred log price interaction), with issuer-cluster bootstrap 95% intervals. "Other industries" are the deals in neither registered regime, including those with no industry code (registered codes). "Net of regime" fits all deals with one common interaction and the financial regime's own intercept, cap slope and price slope; it is a weighted average of the rows above, so the other industries' flatter gradient pulls it down (on the two regimes' deals alone it is +0.11 [0.06, 0.16] held out and +0.12 [0.09, 0.15] in development). Exploratory.

	Pooled held-out			Development		
	n	Median price	Gradient	n	Median price	Gradient
All deals	637	$43M	+0.10 [0.05, 0.15]	2,266	$41M	+0.12 [0.09, 0.15]
Financial (SIC 60–67)	100	$78M	+0.10 [0.004, 0.20]	253	$60M	+0.18 [0.10, 0.25]
Manufacturing and services (SIC 20–39, 70–89)	393	$35M	+0.12 [0.06, 0.17]	1,408	$34M	+0.11 [0.08, 0.14]
Other industries	144	$46M	+0.07 [0.01, 0.17]	605	$56M	+0.10 [0.06, 0.14]
Net of regime	637		+0.09 [0.05, 0.14]	2,266		+0.11 [0.09, 0.14]

held-out finance only just, +0.10 [0.004, 0.20]), and with each regime given its own intercept, cap slope and price slope the pooled interaction is +0.09 [0.05, 0.14], against +0.10 without. Figure 9 draws the result: in both regimes, and in both development and the held-out deals, the line rises with deal size, and the two regimes stay apart at every size. In manufacturing and services the elasticity runs from about 0.1 in the smallest deals to about 0.35–0.45 near $1 billion; in finance from about 0.5–0.6 to about 0.9.

7.5 The drafter

Industry changes which anchor the basket follows more than how large it is. It explains a modest share of the dispersion in levels: adding industry divisions to price and price deciles raises the adjusted R^2 of the basket's share of price by 0.040 on the pool (2–10 points across samples), which the registered rule reads as holding; the first registered read, before industry was taken at filing and the deciles weighted as registered, gave 0.025, read as weak (Appendix A). The filing issuer matters, but less than a raw comparison suggests. On the 966 development records whose issuer

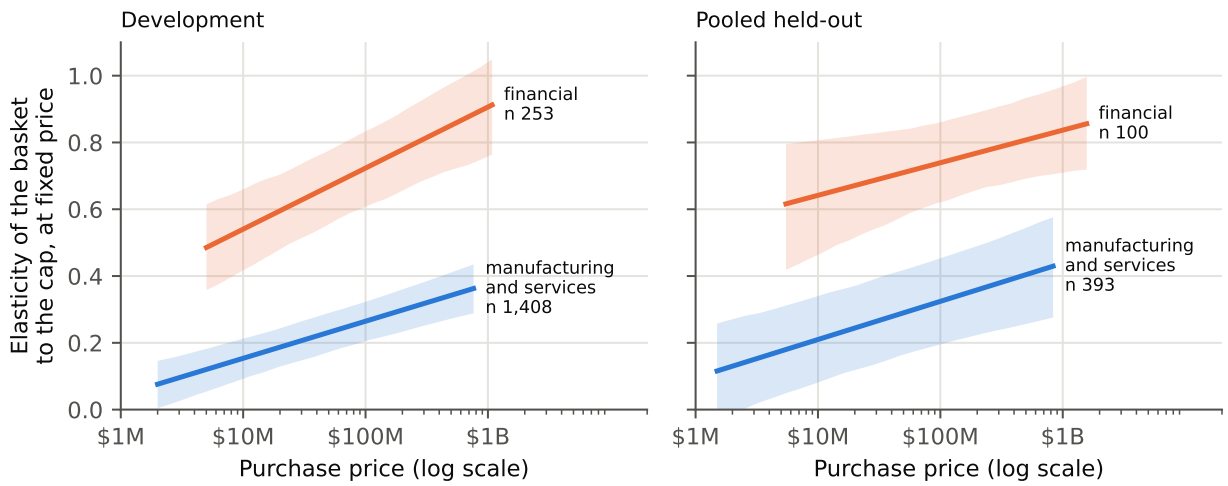


Figure 9: The size gradient. The elasticity of the basket to the cap at fixed price, implied by the interaction model fitted inside each regime, across the middle 90% of each regime's deal prices. Bands are issuer-cluster bootstrap 95% intervals for the curve.

has at least two deals, issuer fixed effects reach an R^2 of 0.63 for the basket's share of price and 0.64 for the cap's, against 0.38 when the same records are assigned to issuers at random; terms cluster by issuer beyond chance. But the fixed effects spend 367 parameters on 966 records. Adjusted for that, they explain about as much as the 72 named covariates (0.40 against 0.36 for the basket, 0.42 against 0.40 for the cap), and they predict a held-out deal worse: the leave-one-out R^2 is -0.07 and -0.05 for issuer identity against 0.31 and 0.36 for the covariates (exploratory). Most issuers here have two or three deals, too few to estimate an issuer's habit well. The filer is also not necessarily the drafter: it is usually the acquirer, and the first draft may come from either side's counsel. The result is consistent with template-driven drafting (Nyarko, 2021) but does not show it.

7.6 Insured deals have lower terms

Where the agreement provides for RWI, the seller's cap is far lower and the basket somewhat lower. In development the insured median basket is 0.50–0.55% of price and the insured cap 1.0–6.2% depending on era, against roughly 0.75% and 10–15% uninsured; in the joint 2020–2025 read, 0.40% and 1.5% against 0.62% and 10.0%. The cap falls by four-fifths or more and the basket by a third to a half, so the terms do not move down together. Uninsured deals after 2020 keep the old levels, so the lower terms go with insurance rather than with the calendar, consistent with Griffith (2019); insured deals may differ in other ways (buyer type, deal size), and this comparison does not separate insurance from them. Whether the co-movement survives inside insured deals is open: no sample has enough insured core deals (31 in development, 7–25 in the test reads) to distinguish an elasticity of 0.15 from 0.75.

8 What reading the context buys

A second registration asked whether the reader's context rules matter, turning them off one at a time over the 47,543 development documents and scoring against two truths: the agreements' own defined terms (1,125 baskets and 457 caps that the agreement names in a parenthetical) and the

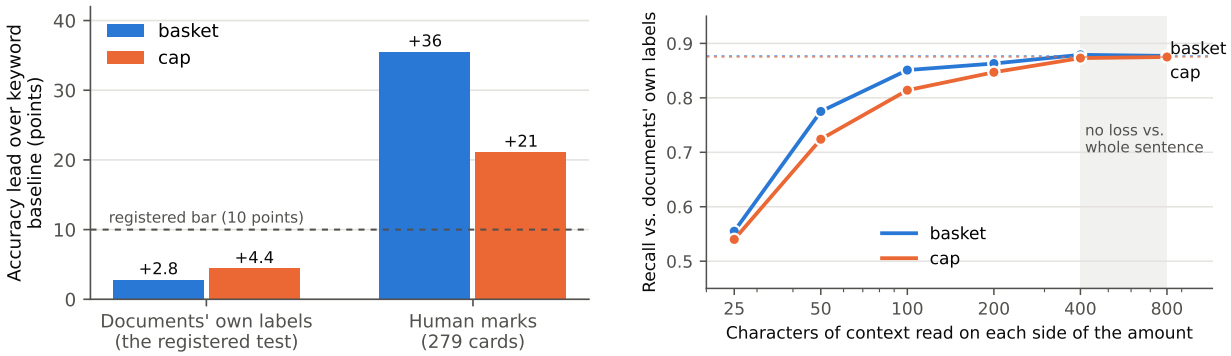


Figure 10: Left: the full reader's accuracy lead over a nearest-keyword baseline, on the agreements' own labels and on the first author's marks. Right: recall when the reader may see only a window of text on each side of the amount; dotted lines are the full-sentence reader.

first author's marks. The full reader was compared with a keyword baseline that gives each amount the role of the nearest role word in its sentence—close to what a generic extractor can do.

Context against keywords. On the agreements' own labels the full reader leads the keyword baseline by 2.8 points on baskets and 4.4 on caps (intervals excluding zero), short of the registered 10-point bar: the registered verdict is *weak*. On the first author's marks the lead is 36 points on baskets and 21 on caps, and 27 and 13 on fresh cards alone (Figure 10). Neither number is the lead on a typical agreement, because each truth is selected. An amount the agreement names in a parenthetical has its keyword right beside it, so the defined-term truth favours the keyword baseline; the registration anticipated that. The marks are biased the other way, and we did not anticipate it: the cards show the full reader's extraction, and where the first author marked an extracted amount correct, that amount is the answer key, so the comparison is scored on the full reader's own successes. The lead on a typical agreement lies between the two, and we cannot say where. The registered verdict is the first number's: weak.

Which rules matter. Three rules carry measurable weight: aggregate-versus-per-claim scope (removing it costs 5.4 points of basket recall), the agreement's own defined terms (adding it alone to a stripped reader gains 5.7), and direction (adding it gains 5.9 points of cap recall). Specific-matter carve-outs and party resolution are small on this truth; the party rule is nearly invisible to it by construction. Cross-reference handling has no measurable effect.

How much text. Reading 400 characters (about 65 words) on each side of an amount loses nothing against the full sentence; 200 characters does not suffice (Figure 10, right). The reader never looks across sentences, which is why it misses consequence clauses that follow the threshold.

The headline under crude extraction. Turning the rules off one at a time and re-estimating, five of the six move the elasticity by about 0.01 or less. One moves it further: without the rule that separates per-claim from aggregate amounts, the elasticity falls by about 0.04 (0.038 [−0.003, 0.077] in a paired comparison), about a tenth of its value, and the shift is clear only in financial deals (0.08 [0.007, 0.17]). That rule also makes the reader more accurate, so the more accurate reader gives the higher estimate. The registered verdicts are that the estimate is robust to removing each rule (the first read found it not robust to the scope rule, by 0.001) and that it does not depend on the context

machinery as a whole; Appendix B explains why the registered comparison is a weak test and gives the paired one in full. What no simple reader changes is the sign, the distance from the null, and the ordering of the two regimes: the financial elasticity stays above the manufacturing-and-services one in all 22 versions of the reader.

9 Discussion

What the results describe. Across deals, the rungs of the ladder are not independent numbers. At fixed price the basket varies with both the price and the cap, and the balance differs by industry: in financial-sector deals the basket varies more with the cap, in manufacturing and services more with the price, and in larger deals more with the cap than in smaller ones. For benchmarking, a basket is best compared with deals of similar cap, price and industry: at the same price, a deal with twice the cap has, on average, a basket about a quarter larger in manufacturing and services and about 70% larger in finance. These are comparisons across deals, not predictions of what a party gains or concedes by negotiating one term. Terms cluster by issuer, which fits the view that they are sticky, carried by templates rather than freshly bargained (Nyarko, 2021), and sits in some tension with the view that they are engineered deal by deal to manage information asymmetry (Gilson, 1984); but issuer identity does not predict a new deal's terms better than the deal's own characteristics, so the data do not decide between the two. Insured deals have far lower caps and somewhat lower baskets; whether the relationship between the terms holds inside them the data cannot yet say.

What they do not. The estimates are associations within a population of dollar-capped private-target deals filed with the SEC: agreements that were signed, not the negotiations behind them or the deals that failed (Section 2.4). They do not identify the effect of exogenously changing one term, and we have no credible instrument for one: the escrow, the obvious candidate, is negotiated with the cap and in about a quarter of escrow deals is the cap. Within-issuer estimates are smaller (0.27–0.39) and exploratory. About a third of agreements with both rungs state a term by reference and sit outside the population, and we do not know whether they follow the same ladder. Adding the deals whose percentage cap our reader can resolve barely moves the estimate, but they are a small slice read by the reader that misses most of the rest, and a better reader that failed its validation leans the other way, within its uncertainty (Appendix C). The question is open. Deals capped at the whole price, about one in nine, cannot enter a regression of the basket on the cap at fixed price at all. The snapshot ignores scheduled caps.

Limitations. Every validation mark is the first author's; no second human marker was available. A blind model read of the same cards, which never saw the extractor's output or the marks, agrees with them on 92% of baskets and 86% of caps. The first author re-read the 20 disagreements inside the core with the two answers shown as A and B, not told which was his. He kept his earlier answer on 12, took the model's on 5 and rejected both on 3; after unblinding, two of the 3 turned out to be the model's answer of no cap, and the third is a set of per-category caps with no single general cap. In every one of the 8 losses the earlier mark had accepted the extractor's value. If the re-read is right, those are extractor errors and precision falls to the re-read rows of Table 2. Nothing independent settles which reading is right: the re-reader is the original marker, and only on the 7 cards where he took the model's answer does a second reader back the re-read. Cards where marker and model agree were not re-read at all. Precision on fresh cards (0.86 for baskets after the re-read) is below the core figure; misses concentrate on percentage-stated terms, outside the core. Outside

the core the reader is much weaker (Section 4.2), and its failures there are systematic—percentages written in words, dollar caps phrased as "in excess of" under an enumerated "in no event", the fundamental-representation ceiling taken for the general cap, and caps that bind only the buyer or one class of claim—and the same errors can seat a misplaced amount as the basket (a basket equal to the escrow in 0.3% of development core records, 0.7% of those with an escrow). The 1995–2020 holdout was read more than once. Two registered mandatory checks on the post-2019 pulls failed (deal-gate rates of 11% and 22% against the registered expectation), reducing yield rather than biasing reads. The archive test is conditioned on agreements that state a threshold term. Insured-deal slopes are underpowered everywhere.

Future work. A fuller measure would replace each nominal term with its schedule— amount as a function of time, representation class and recovery source—and collapse the schedule to an effective cap, the expected ceiling weighted by when claims arrive, so fixed, stepped, stock-linked and tiered caps sit on one axis. The nearest open item is the third of agreements that state a term by reference. Bringing them in needs a cap reader that passes validation; ours handled the wording of these caps but not their structure, and fell short of its bar (Appendix C). Until such a reader exists, our results speak only for deals whose terms are stated in dollars. Beyond that, the insured slope and the mechanism behind the financial-sector regime are open. For basket type, widening a keyword rule's window makes it worse; a model reading the basket's sentence and the next loses almost nothing against the whole region, at about $0.003 per agreement.

10 Conclusion

Among deals whose terms are stated in dollars, at fixed price a larger cap goes with a larger basket, in two regimes that replicate out of sample: in manufacturing and services the basket varies more with the price (cap elasticity about 0.3), in financial-sector deals more with the cap (about 0.8). The pooled elasticity of about 0.5 mixes the two. The relationship grows with deal size. Terms cluster by issuer, but how much of that is a drafter's habit we cannot yet say. Reading the context of each amount decides which number is which; the relationship itself is weakened, not erased, by reading crudely.

Each clause of an acquisition agreement holds a few negotiated points: here the de minimis, basket, cap and survival period of the indemnity. This paper reads those points out of one clause and measures how they move together. It is a first step toward doing the same for the other clauses of an agreement, and toward asking how the points of different clauses depend on one another.

Use of language models. The authors used Claude (Anthropic) to assist with code and with editing the text; the authors take full responsibility for the content.

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A Registered verdicts and deviations

Table 5: Registered hypotheses and verdicts on held-out data.

Claim	Read	Verdict	Estimate
H1 elasticity	Test-A (78)	Not replicated (interval too wide)	0.30 [0.06, 0.55]
	first read (86)	Replicated by rule	0.35 [0.09, 0.57]
	Joint 2020–25 (220)	Replicated	0.49 [0.34, 0.62]
	Archive 1995–2020 (417)	Replicated	0.50 [0.39, 0.60]
H2 five edges	Test-A	Two fail: does not transfer	core triangle holds
	Joint; archive	One fails each	cap–survival; de minimis
S1 industry regimes	Pooled (637)	Holds	+0.54 fin., +0.31 oper.
S2 size gradient	Pooled (637)	Holds	+0.10 [0.05, 0.15]
S3 industry's share	Pooled (637)	Holds	+0.040 adj. R^2
	first read (673)	Weak	+0.025 adj. R^2
Insured slope	Every read	Underpowered	–
v2 extractor sensitivity	All reads	Robust (first read); moot now	core 759 → 762
Context: rules matter (R1)	Development	Scope, terms, direction	see §8
Context vs keywords (R2)	Development	Weak (labels); large (marks)	+3/+4; +36/+21 pts
Sufficient window (R3)	Development	400 characters	–
Headline under ablation (R4)	Development	Robust (first read: not, by 0.001)	0.373 vs [0.355, 0.459]
		Does not depend on context machinery	keyword 0.360 [0.292, 0.437] covers 0.411

Deviations, in full. (1) The post-2019 split's first draw was discarded before any outcome existed and the unit of assignment amended. (2) Mandatory gate-rate checks failed on two pulls. (3) The 1995–2020 holdout was re-read after development-justified rule changes. (4) A crash in the partial-correlation code on test-A was fixed by replacing an exact solve with least squares before the read. (5) The percentage-basket reader was designed after the test reads; its effect was registered and read once, and the extractor frozen (Appendix C). (6) The registered pooled read counted test-A twice (759 rows, 673 deals); it now pools the joint read and the archive test, each agreement once. (7) Core precision was first reported as 0.98 without noting it was in-sample. (8) The ablation scorer was fixed before the read to report, rather than compute, a partial correlation with fewer than 30 complete records. (9) After the extractor was frozen, a percentage-cap reader was built on the first author's instruction, reversing an earlier decision to make no further extractor change. It was designed on 25 in-sample cards; its pass bar sat in a draft registration that was never adopted, and it failed that bar on fresh cards. One mark was corrected after the result was known. Nothing in the paper's estimates uses it (Appendix C). (10) The blind re-read of first-author/model disagreements was added after the estimates were read, in response to a critical review; it changes the precision reported, not the estimates. (11) After all reads, the truncation null was re-specified as a conditional simulation (Section 6); it is 0.11–0.13 rather than the registered 0.08. Against it test-A's interval did not exclude the null even in the first read. No other verdict changes. The ablation re-read by regime (Section 8) is also post hoc; the registered verdict on the pooled estimate stands. (12) The registered ablation rule (R4) compares an arm's point estimate with the full reader's interval. Because the arms share most of their deals, this is not a test of their difference, and its verdict at the margin depends on the bootstrap seed. The paired comparison in Section 8 was added after the registered read to measure the shifts directly; the registered verdicts are reported unchanged. (13) A code review after every read (amendment of 2 November 2026, registered before any re-run) found faults in the extractor, the population rules and the held-out screening, and every sample was re-read once with them fixed. Each held-out sample is now screened against development and against the other held-out samples, so no agreement is read twice; the post-2019 pull was completed (Section 4), so the joint read is a second reading of an enlarged test-B; industry is the code assigned at filing, not today's; the pooled gradient and industry-share statistics follow their registration (each sample's own weighted median; weighted deciles); validation is post-stratified on today's core. Where a verdict changed, the table gives both readings: test-A no longer meets the registered rule (0.30 [0.06, 0.55] on 78 deals; the first read's 86 included copies of agreements development already had), S3 moves from weak to holds, and the ablation's R4 verdict moves from not robust (by 0.001) to robust. The amendment named the S3 change in advance.

B Technical details

The price check. A blind hand check of 50 deals drawn at random from the held-out core (the first author marked the price from the same excerpts, without the label) agreed exactly on 41 of the 44 both priced, and on re-reading the 3 disagreements were the marker's misses (a stock component, a non-cash component, a closing payment taken for the total); of the 6 the marker left unpriced, 3 labels are wrong (one counts a note twice, one prices a half-interest, one cannot be traced). The reliability of log price is 0.996, and correcting for measurement error of twice the measured variance moves the pooled held-out cap coefficient from 0.50 to 0.49 (exploratory).

Error in the rungs. Error in log cap attenuates β, and a misassigned rung—an escrow or a special-matter cap read as the general cap, a de minimis read as the basket—can put correlated

error in both sides of the regression. Its size is measured on the hand-checked core cards (Section 5; 182 with a decided truth, weighted to the core, the blind re-read's corrections applied). Of these, 173 have both true rungs in the dollar population and 9 do not: the extractor admitted a deal whose true cap is a special cap or an escrow, or whose true basket is a de minimis. On the 173, the extracted and true values give β within about 0.03 of each other (true minus extracted +0.025 [−0.01, 0.07]); value errors are rare (4 baskets, 5 caps), their weighted variances are 0.004 (cap) and 0.008 (basket) in $\log_{10}$ units, and their covariance is zero. A classical errors-in-variables correction with these moments raises β by about 0.01 in every sample (pooled 0.50 to 0.51; manufacturing and services 0.31 to 0.32), and by 0.04 at four times the measured variance. Counting the 9 wrongly admitted deals as well (extracted values on all 182 cards against true values on the 173), true minus extracted is +0.10 [−0.09, 0.26]: if anything, extraction error biases β down, toward the null, not up (exploratory).

The truncation null. The core requires $\$1{,}000 \le B < C \le P$. The bounds on C and P select on regressors and leave the conditional mean of $\log B$ intact; the bound $B < C$ truncates the outcome at a regressor and by itself produces a positive β. The null that measures this holds log basket independent of log cap given log price. Each simulated sample keeps every deal's observed cap, price and weight and redraws its basket from a marginal, rejecting draws outside [\$1,000, C), until every deal has one; the regression is refitted on 200 such samples. Three marginals are used. In levels, $\log B = a + g \log P + e$ with normal e, the parameters taken either from the regression of log basket on log price in the observed sample (N1) or from a truncated-normal fit that undoes the truncation (N2). In shares, the log basket share $\log(B/P)$ is drawn from its empirical distribution in the sample (N3), which fixes the price slope at one. At fixed price, independence in levels and in shares is the same hypothesis; the variants differ only in the basket's marginal. In development the null is 0.11 [0.07, 0.14] (N1), 0.12 [0.08, 0.15] (N2) and 0.11 [0.07, 0.15] (N3); on the pooled held-out deals 0.11–0.13, with upper bounds of 0.17–0.20; in manufacturing and services 0.07–0.08 and in the financial regime 0.16–0.22, which are wide on 100 deals; on test-A's 78 deals 0.07–0.17, with upper bounds of 0.24–0.34. The registered null, 0.08 [0.02, 0.14], came from an earlier construction that drew basket and cap independently given price and kept the admissible pairs (N4; 0.06 in development). That changes the sample's composition, and the conditional nulls are the better benchmark. Test-A's interval reaches the registered 0.08 and the conditional nulls N1–N3; only N4 on test-A (0.02) lies below it. Every other sample's interval excludes every null (Section 7).

Why not a truncated-regression estimator. The null is a benchmark for testing, not a bias to subtract. Under independence much of the basket's latent distribution lies above the cap, so the truncation binds; in the data 1–2% of baskets lie within a factor of two of their cap, so it rarely binds. A truncated-normal maximum-likelihood estimator, which models the bound [\$1,000, C) on the outcome directly, confirms this. On the null samples it returns 0.00. On the data it gives 0.38 [0.33, 0.43] in development, 0.49 [0.40, 0.57] on the pooled held-out deals, 0.78 [0.66, 0.90] in the financial regime and 0.29 [0.18, 0.40] in manufacturing and services: within 0.03 of least squares in every sample (exploratory). We report least squares, the registered estimator, which does not need the truncated estimator's assumption of normal errors.

A language-model baseline. We gave Claude Sonnet 5.5 the same indemnity region and labelled price and asked for the basket, the cap and the basket type as structured output (304 agreements, \$4.21). Scored against the marks (Table 6), the rules are slightly ahead on amounts in the core, the model is ahead on agreements where the rules missed a rung, and the model is

Table 6: Rules against a language-model baseline, scored on the first author's marks. Core rows: stratum-weighted as drawn, and post-stratified on the current core. The measure is the correct value, or correctly none, so the rules' figures differ from the precision in Table 2. Basket type: agreement with the first author where both commit to tipping or deductible.

	Rules	Claude Sonnet 5.5
Basket amount, core (stratum-weighted)	0.971	0.966
Cap amount, core (stratum-weighted)	0.967	0.938
Basket amount, core (post-stratified, 188 cards)	0.966	0.961
Cap amount, core (post-stratified, 188 cards)	0.983	0.943
Basket, agreements with a missed rung (50)	0.82	0.88
Cap, agreements with a missed rung (50)	0.66	0.72
Basket type, agreement with first author	0.81 (156 calls)	0.95 (223 calls)

much better at basket type. Cards the original marks could not score (a rung marked wrong with no value typed, or marked absent where the rules had bound one) were re-marked by the first author for this comparison, so no card is dropped. The comparison still favours the rules in two ways we cannot remove: where the first author marked an extracted value correct, that value is the answer key, and the model was not given the labelled escrow amount that the rules use to resolve escrow-based caps.

Basket type. Whether a basket tips or deducts is stated in the consequence clause, often in the sentence *after* the threshold. The rules read one sentence and misread "liable for all such Losses in excess of $X" as tipping. On the 39 cards where any two readers disagreed, the clause text supported the model in every clear case; the first author's own errors concentrated on "only to the extent that Losses exceed $X", which is a deductible. A re-run with an explicit "silent" class (a threshold with no stated consequence; $4.26) agreed with the first author on 96% of committed calls and matched a careful reading of the text on 11 of 12 disputed clauses.

Is the model's advantage more text or better reading? An exploratory grid crossed the reader with the window—the basket's sentence, that sentence and the next, the whole region—on the 241 marked cards with a basket ($1.26). Given only the sentence the rules read, the model agreed with the first author on 96% of committed calls against the rules' 82%, and corrected 24 of the rules' 30 wrong calls. Widening the window *lowered* the rules to 74%, because more text brings more stray cues from carve-outs and other rungs; it left the model at 96% and bought only coverage ("silent" calls fell from 20 to 8 with one more sentence). The difference is reading, not context. A second check generated agreements from the first author's marked terms with interchangeable phrasings modelled on filings and asked each reader to recover the terms (240 agreements; $1.66): the model recovered the full meaning—amounts and type—in 99%, the rules in 49%. The rules' failures are specific wordings: "responsible for only the amount in excess of $X" (never typed), deductible language recognised only when the basket is named "Deductible", consequences stated in a second sentence, and caps written "fifteen percent (15%) of the Purchase Price". These agreements are cleaner than filings, so 99% is a ceiling. The model's labels, read from the basket's sentence and the next with silent and unclear clauses as their own class, are used only in this comparison. Where basket type enters the analysis (the template weights and the drafter covariates of Section 7.5), it is the rules' label, which agrees with the first author 81% of the time.

The headline under crude extraction. The registration asked two questions of the development elasticity. First, is it robust to any single rule: does every reader with one rule removed give

an elasticity inside the full reader's interval? It is. With each rule removed in turn the elasticity stays at 0.40–0.42, except that removing scope lowers it to 0.373, inside the full reader's interval (0.411 [0.355, 0.459]). The registered verdict is *robust*; the first read, before the code review, put the scope arm at 0.364, 0.001 below an interval of [0.3645, 0.461], and read *not robust*. Second, does the headline depend on the context machinery at all? The registered test is whether the keyword baseline's interval covers the full reader's estimate. It does: the baseline gives 0.360 [0.292, 0.437] on a third fewer deals, an interval that contains 0.411, so the registered reading is that the headline does not depend on the context machinery.

What the registered test can and cannot show. Both registered comparisons set one reader's point estimate against another's interval. The readers share 95–100% of their deals (the keyword baseline 58%), so their estimates are strongly correlated, and that comparison is not a test of whether they differ. It is also unstable at its edge: re-running the full reader's bootstrap with 20 different seeds puts the lower bound anywhere from 0.355 to 0.367, a spread larger than the margin by which the first read failed. A paired test—each bootstrap resample of issuer clusters refitted under both readers—measures the shift directly (exploratory). Four of the six rules have no detectable effect: each shifts the estimate by at most 0.010 and every interval includes zero. Party resolution shifts it by −0.012 [−0.023, −0.001], detectable but negligible. Scope shifts it most: removing it lowers the elasticity by 0.038 [−0.003, 0.077], about a tenth of its value, and by 0.075 on the deals both readers admit, so the rule changes the amounts read rather than which deals enter; pooled, the interval reaches zero. The keyword baseline's shift, −0.051 [−0.117, 0.011], is about as large but too imprecise to separate from zero; its registered pass reflects its wide interval as much as its agreement. The dependence is on a rule that improves accuracy: removing scope costs 5.4 points of basket recall, so the more accurate reader gives the higher estimate. What does not depend on the rules is the sign, the distance from the null and the ordering of the regimes.

By regime. Paired within each regime, the manufacturing-and-services estimate does not move detectably under any single removal or the keyword baseline (shifts of at most 0.03, every interval but one including zero; cross-references, +0.005, just excludes it). The financial estimate does: removing scope lowers it by 0.082 [0.007, 0.171] and removing the specific-matter rule by 0.068 [0.017, 0.124], from 0.72 on 253 deals; why these two rules matter more in financial agreements is not measured here. Under the registered yardstick both regimes would read as robust, because the financial interval is wide; the paired test is the better guide. The ordering survives every reader: the financial coefficient stays above the manufacturing-and-services one in all 22 arms. Stripping the reader down further moves both estimates: with all rules off the financial coefficient falls to 0.50 and the other rises to 0.35, and with 100 characters of context or fewer they are 0.45–0.59 and 0.28–0.40. A crude reader blurs the two regimes toward each other.

C Terms stated by reference

Nothing in the main text uses what follows. It is here so that a reader can judge the excluded deals, and because deviations (5) and (9) refer to it.

Percentage baskets. A revised reader that resolves a basket stated as a percentage of the price recovered 218 baskets on priced records, 23 of 25 correct on fresh marked cards. None enters the core, which admits only baskets stated in dollars. Few could have, because deals that state the basket as a percentage mostly state the cap by reference too: in the first read, which admitted

them, 22 entered in development and 3 across the held-out reads, and re-reading every registered claim with them added, under a registration made first, changed no verdict and no estimate beyond the second decimal.

Caps stated as a percentage in digits. The frozen reader resolves "10% of the Purchase Price". Adding those deals (98 in development, 51 held out) moves the cap coefficient by at most 0.02 (0.41 to 0.39; 0.49 to 0.47). The difference between the two groups' slopes spans zero in both samples and changes sign between them (-0.10 $[-0.52, 0.21]$ in development, $+0.09$ $[-0.54, 0.45]$ held out). The intervals are too wide to show the slopes are equal.

A reader for the rest, and why we do not use it. The frozen reader misses percentages written in words ("ten percent (10%)", "one-third"), and it often takes a fundamental-representation ceiling at the whole price for the general cap. After the main results were known we built a revised reader for these cases, designed on 25 development agreements the first author had marked. Before using it we set a bar: at least 20 of 25 right on fresh development cards drawn at random from the 340 records whose cap it changed, marked blind by the first author with neither reader's answer shown. It got 18 of 24 (one card, model-assisted, excluded). Re-reading the disputed cards after the result was known showed one of the first author's marks was wrong—a cap of 5% plus a separate 5% for one class of claim, where the general cap is the first 5%—and correcting it gives 19 of 24. Both fall short. The misses were systematic: the reader dropped caps stated as the escrow or as a share of the seller's promissory notes, took the whole price where the escrow capped, and took one rung of a split cap or a carve-out ceiling for the general cap.

On development data the failed reader grows the percentage-capped group from 98 to 270 deals, and in that read the percentage-capped deals' basket keys to the cap a little less than the dollar-capped deals' does (slope difference -0.11, 95% interval -0.30 to 0.07). In the first read, before percentage-stated baskets were taken out of the core, the difference was -0.21 $[-0.39, -0.02]$. Neither settles whether leaving these deals out is harmless: both come from a reader that failed validation, on development data, with no held-out read.